

I: Infinite Go

This chapter it's a draft, at least for now. I'll include my thought process and most of what I'm able to understand/formalize about it.

To begin with, let's see *where* we are playing now. Our infinite board should be able to have also infinite dimensions and every one of those could be or not infinite as well.

Let's begin with the simplest case (I hope), let's play on a two dimensional board that has one infinite dimension.

Let G be a go board such that,

$$G = \{1, 2, \dots, n\} \times \mathbb{Z}.$$

for $n \in \mathbb{N}$. We'll use $\mathbb{Z}$ for simplicity as it's a better (easier) representation of the infinite board. As the number of players does not matter at all, let's use only two players and, the canonical adjacency. Our game will be $\Gamma = (G, E, A_{ort}, \Delta_{Go_\infty})$. At least for now, we are not sure how Δ_{Go_∞} works but, we will get there.

The very first change in this new game or extention of one its the next. A player in the infinite game of Go its allowed to play a line of stones or pass. a line of stone might be or not infinite (that correspond to the player to choose). This is neccesary to be able because if not, the difference between a really big finite board and a infinite one will reaming the same for almost the whole game. Being able to play infinite groups its a necessary change in the dynamic of the game. You might say, what about intersecting infinite lines? Very simple, as shown in the right board of Figure 1.1, a new line of stones that intersect another just splits, when the line touches a point that has a color that its not 0, then the color that was just remanins, if the color its 0 then it changes into whatever color the player might be.

Our board could (and for us, will) look like this, Figure 1.1:

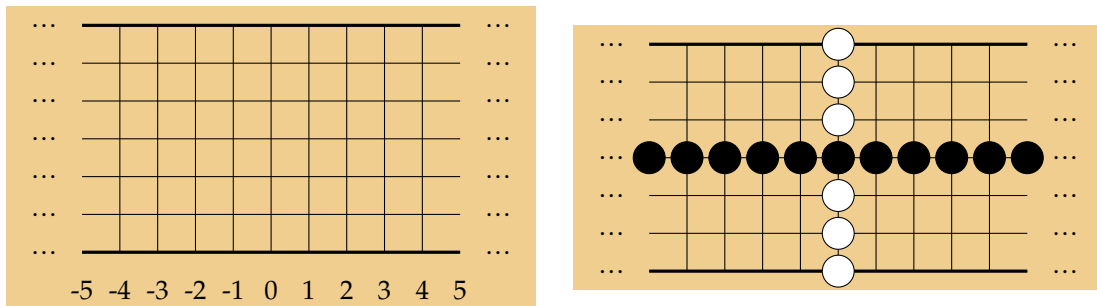


Figure 1.1 – A $7 \times \infty$ board and two intersecting plays that resolves by definition.

Notice that, if the white line were infinite, such play generated two infinite lines in one turn, which is crazy.

The very first problem on the infinite board it's how to determine a capture. Any other prior definition should work just fine.

Let's say we are in the next position Figure 1.2

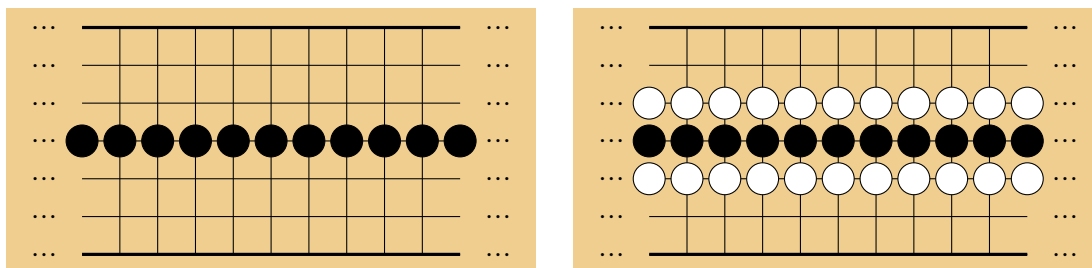


Figure 1.2 – An infinite board with an infinite black group being capture.

As this group is infinite (numerable), will touch the very end of the board so we can capture it by surrounding it at the top and bottom.

We can be sure that this is capture because if the black group it's I , then $K(I) \subset S$ where S is the set of the two white groups. That is, $L(I)$ should be 0. Now let's take a look at another example, an infinite group but only on one side.

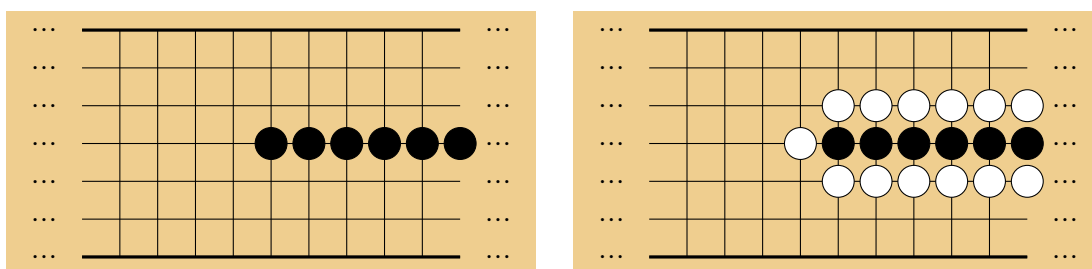


Figure 1.3 – An infinite, on one side, black group being capture.

In this case, not using the whole board, even though it's infinite, its better. As it provides one more liberty (or does not loses it).

Let's see the following example, we might into the nature of inifinite Go.

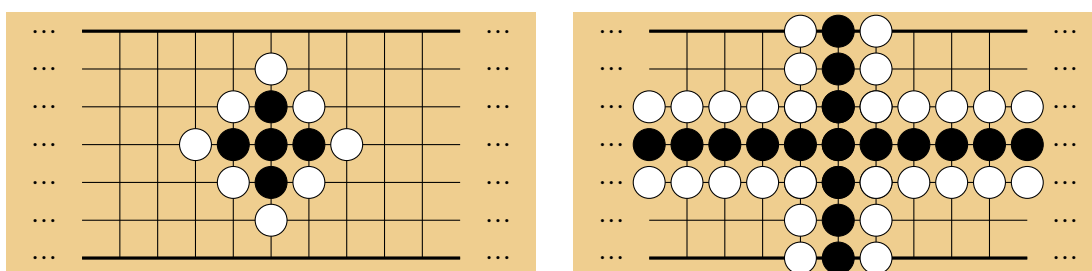


Figure 1.4 – Two captures.

It seems that finite groups are stronger than infinite ones as, again, they don't lose liberties due to not being infinite.

1.1 SuperKo at infinity

We cannot use Zobrits method for the infinite Go as we will need infinite many bits. We need a better method, let's try to make one.

If we remember the nature of the SuperKo rule, it's says that we cannot replicate a prior board via a stone play. Again, we are using this version of SuperKo (Positional).

Because of the rule, every game has to be a line, that is, as we cannot go back into the history of plays (at least it's a pass but will not take in consideration those). From the empty board

to the very end, we have a straight forward path. Not even that but, because of the rule, the whole game of Go its a *tree*. That is, starting from the empty board, there are 362 possible plays (361 stone placements and a pass) and we cannot go back. From those, every possible outcome, that is, any end of a match can be found by following a line of unrepeated boards.

Let's define a way to measure how close are two given boards are.

First, lets notice that, in the very beggining, there are exactly one region and not much else in the board. That is, before the first turn, $|J| = 1$ (the big empty region). Lets say that in the infinite Go you can play a stone or you can play a line of stone (infinite or not) and, of course, you can pass. Yes, I'll admit a new move, in the infinite Go a player can play a line of stones (straight), not matter if finite or not.

Now, the very first turn will make so $|J| = 1$ if the black player plays a stone or a finite line. If the player chooses to play an infinite one, then $|J| = 2$ unless the play plays at infinity (I have no idea how). Lets pretend it did not play at infinity, we have just played an infinite group, with that zobrist method or any method that uses the stone itself go nuts and can't be use here but, $|J|$ it just a little bigger, finitely bigger. Now, this is not perfect as there are infinite many ways to play an infinite group so we end up with $|J| = 2$. Before going forward, lets see an example.

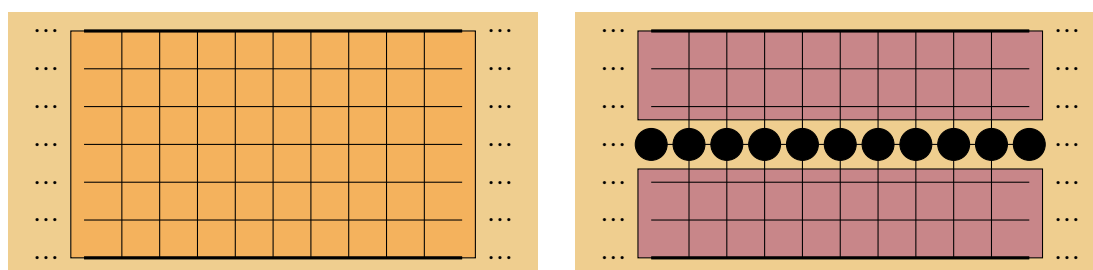


Figure 1.5 – A board where the number of regions (neutral and black) merely grows by one when an infinite group is played.

As I said early, playing such group vertically (in this case will be finite but you get the idea) or shifting it up or down, will end up with the same number of regions.

The first problem is to determine if the number of regions are always finite or not.

The second problem is how to determine those boards that has the same number of regions are infact different, we cannot look at them as that's too costly. My idea is, if we can solve the first problem, then we can measure every region and, by that, we can tell even if two boards have the same number of regions, those regions might not be equal (by biggest and smallest radius or something like that).

The third problem its color shifting, two boards might not be equal but the regions could be exactly the same (number and size), but the colors of the regions surrounding them might not. I have not even the slightest idea on how attack this.

Lets attack the first problem.

Given an infinite game of Go, the number of regions are always finite? If there are infinite moves then the answer is no.

In order to see this, lets consider the infinite one dimensional Go.

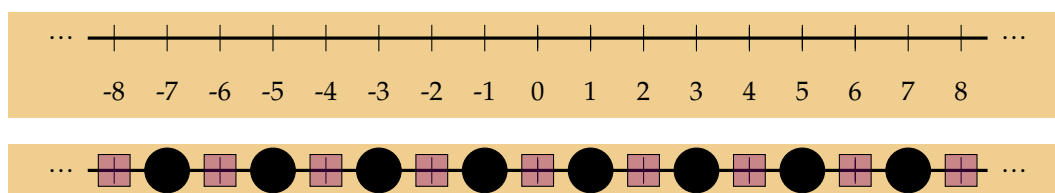


Figure 1.6 – A one dimensional infinite Go board and a example of infinite many regions by infinite many plays.

Is it possible to do many examples like this that end up with infinite many regions. The trick here, and perhaps my salvation, is that there are infinite plays, which is, given the nature of the game, something that shouldn't happen. If we restrict ourselves into only finite many plays, let's see if our problem is solvable. The question now goes as:

Given an infinite game of Go, the number of regions generated by a finite number of plays are always finite too?

I'll give a shot via induction.

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